



Universidade do Minho
Escola de Engenharia
Departamento de Informática

KNIME

Titanic Dataset

MESTRADO EM ENGENHARIA DE TELECOMUNICAÇÕES E
INFORMÁTICA
MESTRADO integrado EM ENGENHARIA DE TELECOMUNICAÇÕES
E INFORMÁTICA
Inteligência Artificial para as Telecomunicações

Objectives

- The primary goal of this task is to **create a comprehensive and efficient workflow** tailored specifically for pre-processing the Titanic dataset.
- The final result will be used to train a **Clustering** model, based on the **K-Means** algorithm.

Dataset

Titanic

- A.csv file: This file contains information about the passengers aboard the Titanic, which sank after hitting an iceberg in 1912.

How to download

- Access <https://github.com/datasciencedojo/datasets/blob/master/titanic.csv>
- Click Download.

Dataset - Download

[datasets](#) / [titanic.csv](#)

...



Phuc H Duong changed name of titanic

4cd38e7 · 9 years ago

History

Preview

Code

Blame

892 lines (892 loc) · 58.9 KB

Raw



Search this file

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fa
1										
2	1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.
3	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	7.

Dataset- Features

12 attributes

- **PassengerId:** An identification number for each passenger;
- **Survived:** This is the target variable (0 = The passenger did not survive, 1 = The passenger survived);
- **Pclass:** Passenger class (1 = Upper class, 2 = Middle class, 3 = Lower class);
- **Name:** Full name of the passenger;
- **Sex:** Gender of the passenger (male or female);
- **Age:** Age of the passenger in years (with some missing values);
- **SbiSp:** Number of siblings/spouses aboard the Titanic;
- **Parch:** Number of parents/children aboard the Titanic;
- **Ticket:** Ticket number;
- **Fare:** The ticket fare the passenger paid;
- **Cabin:** Cabin number (with many missing values);
- **Embarked:** Port of embarkation (with C = Cherbourg, Q = Queenstown, S = Southampton).

Data Readers



Settings Transformation Advanced Settings Limit Rows Encoding Flow Variables Job Manager Selection Memory Policy

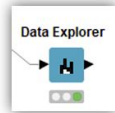
Input location

Read from Local File System

Mode ☒ File ☐ Files in folder

File C:\Users\Ana Sousa\Downloads\titanic (1).csv Browse...

Data Preparation – Data Explorer



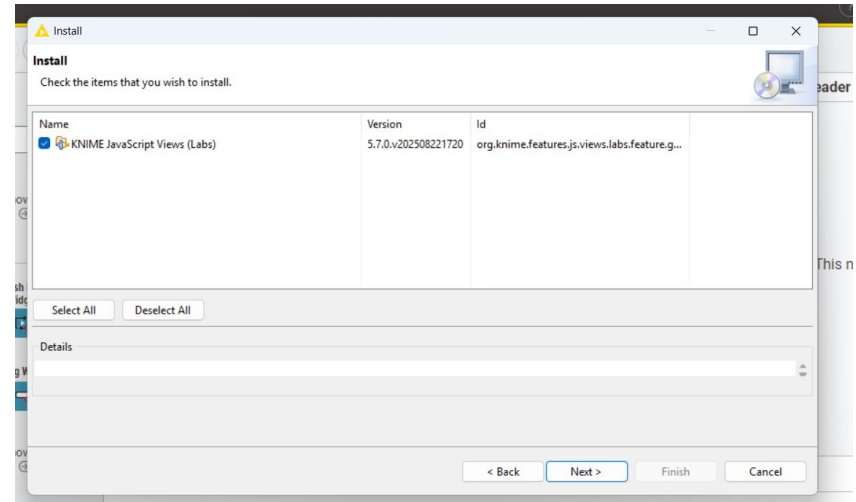
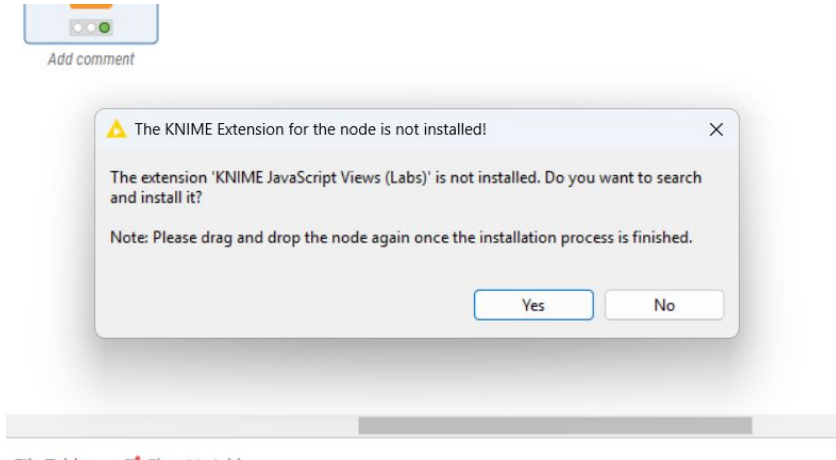
Install **KNIME**
 JavaScript Views (Labs)
 extension

Column	Exclude Column	Minimum	Maximum	Mean	Standard Deviation	Variance	Skewness
+ PassengerId	☐	1	891	446	257.354	66231	0
+ Survived	☐	0	1	0.384	0.487	0.237	0.479
+ Pclass	☐	1	3	2.309	0.836	0.699	-0.631
+ Age	☐	0.420	80	29.699	14.526	211.019	0.389
+ SibSp	☐	0	8	0.523	1.103	1.216	3.695

- Go to [KNIME Community Hub](https://www.knime.com/community-hub);
- Search for “Data Explorer”;
- Drag and drop into the workflow editor;
- Install the extension.



Data Preparation – Data Explorer



Data Preparation – Data Explorer

- Connect the output port of the csv reader node with the input port of the data explorer node.
- Execute the node.
- Click on the magnifying class to check the output.

Data Explorer View

Numeric Nominal Data Preview

Search:

Column	Exclude Column	Minimum	Maximum	Mean	Standard Deviation	Variance
 PassengerId	☐	1	891	446	257.3538420152301	66231
 Survived	☐	0	1	0.3838383838383839	0.48659245426485753	0.23677221654749744
 Pclass	☐	1	3	2.3086419753086447	0.8360712409770491	0.6990151199889029
 Age	☐	0.42	80	29.69911764705884	14.526497332334035	211.01912474630785
 SibSp	☐	0	8	0.5230078563411893	1.1027434322934313	1.2160430774662976
 Parch	☐	0	6	0.3815937149270483	0.8060572211299488	0.6497282437357351
 Fare	☐	0	512.3292	32.20420796857465	49.69342859718089	2469.4368457431156

Showing 1 to 7 of 7 entries

Data Preparation – Data Explorer

- Connect the output port of the csv reader node with the input port of the data explorer node.
- Execute the node.
- Click on the magnifying class to check the output.

Data Explorer View

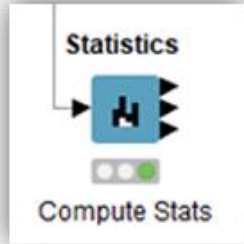
Numeric Nominal Data Preview

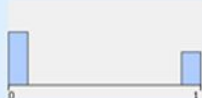
Search:

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 PassengerId	☐	1	891	446	257.3538420152301	66231
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 Pclass	☐	1	3	2.3086419753086447	0.8360712409770491	0.6990151199889029
 Age	☐	0.42	80	29.69911764705884	14.526497332334035	211.01912474630785
 SibSp	☐	0	8	0.5230078563411893	1.1027434322934313	1.2160430774662976
 Parch	☐	0	6	0.3815937149270483	0.8060572211299488	0.6497282437357351
 Fare	☐	0	512.3292	32.20420796857465	49.69342859718089	2469.4368457431156

Showing 1 to 7 of 7 entries

Data Preparation – Statistics

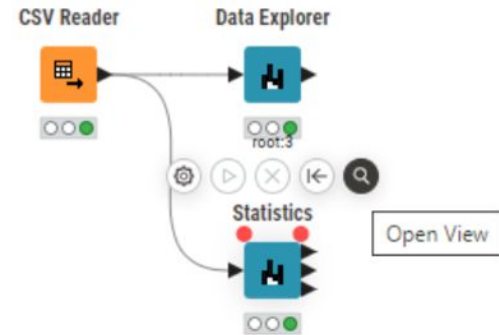


Column	Min	Mean	Median	Max	Std. Dev.	Skewness	Kurtosis	No. Missing	No. +∞	No. -∞	Histogram
PassengerId	1	446	?	891	257,3538	0.0	-1,2	0	0	0	
Survived	0.0	0,3838	?	1	0,4866	0,4785	-1,775	0	0	0	
Pclass	1	2,3086	?	3	0,8361	-0,6305	-1,28	0	0	0	
Age	0,42	29,6991	?	80	14,5265	0,3891	0,1783	177	0	0	

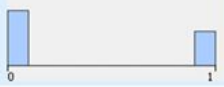
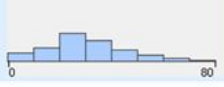
- Offers a range of statistical measures, such as mean, median, standard deviation, minimum, maximum, quantiles, and more.

Data Preparation – Statistics

- In the Node Repository, search for the Statistics node and drag into the workflow editor.
- Connect the output port of the CSV Reader node to the input port of the Statistics node.
- Hover over the Statistics node and click on its magnifier icon.

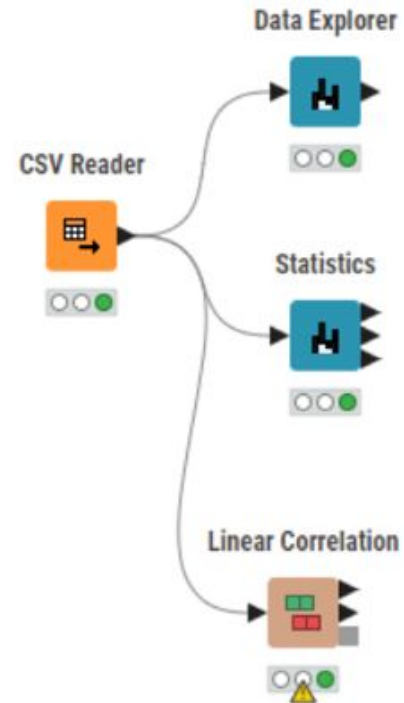


Data Preparation – Statistics

Column	Min	Mean	Median	Max	Std. Dev.	Skewness	Kurtosis	No. Missing	No. +∞	No. -∞	Histogram
PassengerId	1	446	?	891	257,3538	0.0	-1,2	0	0	0	
Survived	0.0	0,3838	?	1	0,4866	0,4785	-1,775	0	0	0	
Pclass	1	2,3086	?	3	0,8361	-0,6305	-1,28	0	0	0	
Age	0,42	29,6991	?	80	14,5265	0,3891	0,1783	177	0	0	
SibSp	0.0	0,523	?	8	1,1027	3,6954	17,8804	0	0	0	

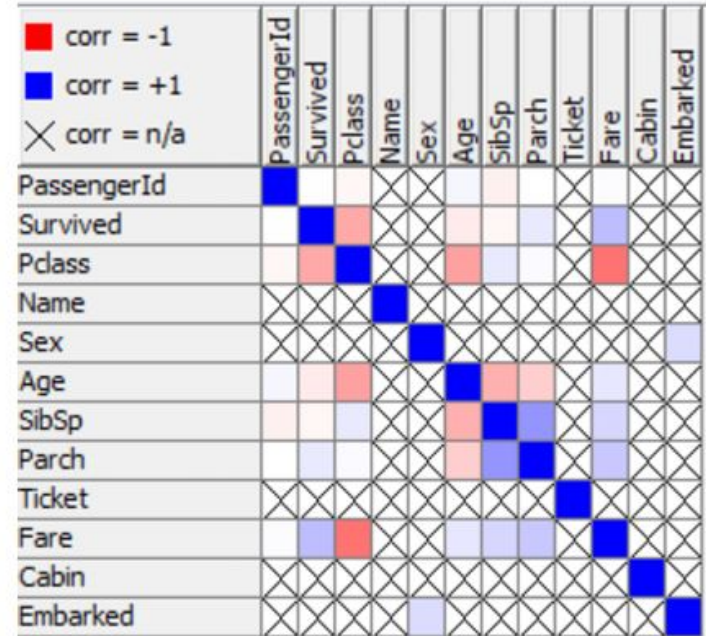
Data Preparation – Linear Correlation

- In the Node Repository, search for the Linear Correlation node and drag into the workflow editor.
- Connect the output port of the CSV Reader node to the input port of the Linear Correlation node.
- Hover over the node and click on its magnifier icon.



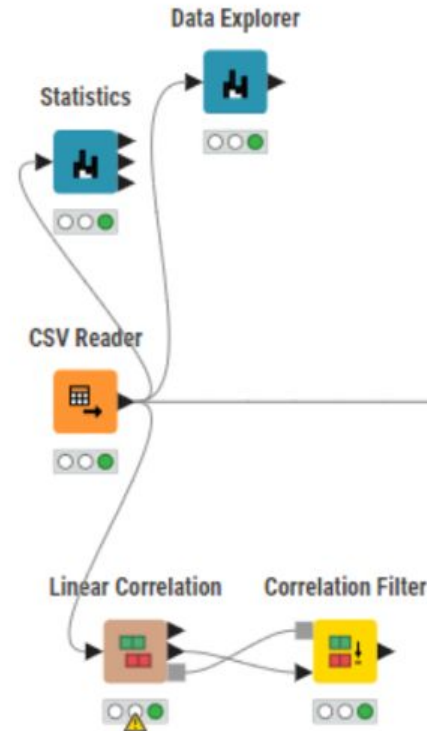
Data Preparation – Linear Correlation

- A correlation coefficient close to 1 indicates a strong positive linear relationship, while a coefficient close to -1 indicates a strong negative linear relationship. A coefficient close to 0 suggests little to no linear relationship between the variables.



Data Preparation – Correlation

- Search for the Correlation Filter node and drag it into the workflow editor.
- Connect both the output ports of the Linear Correlation node to the input ports of the Correlation Filter node.
- Execute the node and inspect its output on the node monitor below.



Data Preparation – Correlation

Rows: 12 | Columns: 12

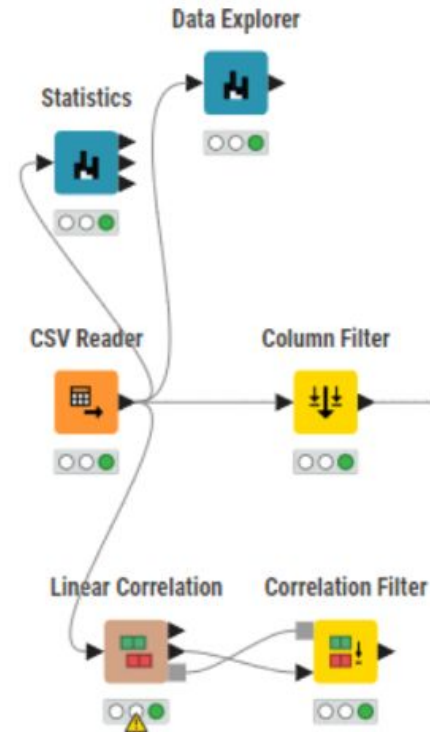
Table  Statistics 

#	RowID	Passenge... Number (dou...	Survived Number (dou...	Pclass Number (dou...	Name Number (dou...	Sex Number (dou...	Age Number (dou...	SibSp Number (dou...	
1	Pass...	1	-0.005	-0.035	?	?	0.037	-0.058	-0.0
2	Survi...	-0.005	1	-0.338	?	?	-0.077	-0.035	0.0
3	Pclass	-0.035	-0.338	1	?	?	-0.369	0.083	0.0
4	Name	?	?	?	1	?	?	?	?
5	Sex	?	?	?	?	1	?	?	?
6	Age	0.037	-0.077	-0.369	?	?	1	-0.308	-0.1
7	SibSp	-0.058	-0.035	0.083	?	?	-0.308	1	0.4
8	Parch	-0.002	0.082	0.018	?	?	-0.189	0.415	1

- Here you can see the ? symbol due to the presence of the missing values.

Data Preparation – Filtering

- Features considered irrelevant must be excluded.
- In the Node Repository, search for the Column Filter node. Drag and drop it into the Workflow Editor. Connect the output port of the CSV Reader node to the input port of the Column Filter node.
- Open the Column Filter node's configuration panel.



Data Preparation – Filtering

- What set of features can be considered irrelevant for clustering in this case?

-PassengerID
-Name
-Ticket
-Cabin;
-Embarked.

Column filter

Manual Wildcard Regex Type

Search Aa

Excludes

- Passengerid
- Name
- Ticket
- Cabin
- Embarked

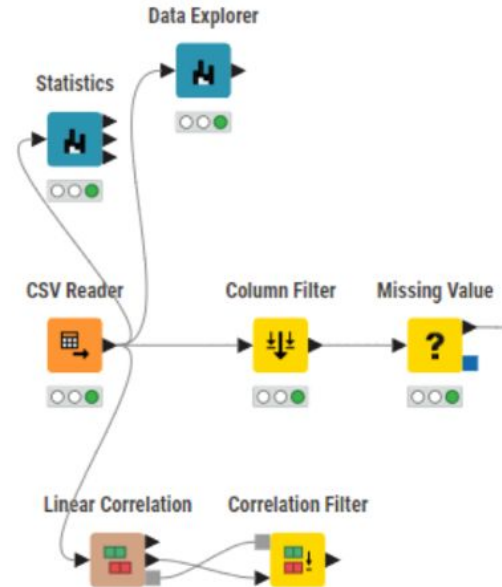
Includes

- Survived
- Pclass
- Sex
- Age
- SibSp
- Parch

Any unknown columns

Data Preparation – Missing Values

- In the Node Repository, search for the Missing Value node. Drag and drop it into the Workflow Editor. Connect the output port of the Column Filter node to the input port of the Missing Value node.
- Replace the missing values of the attribute Age by its median.



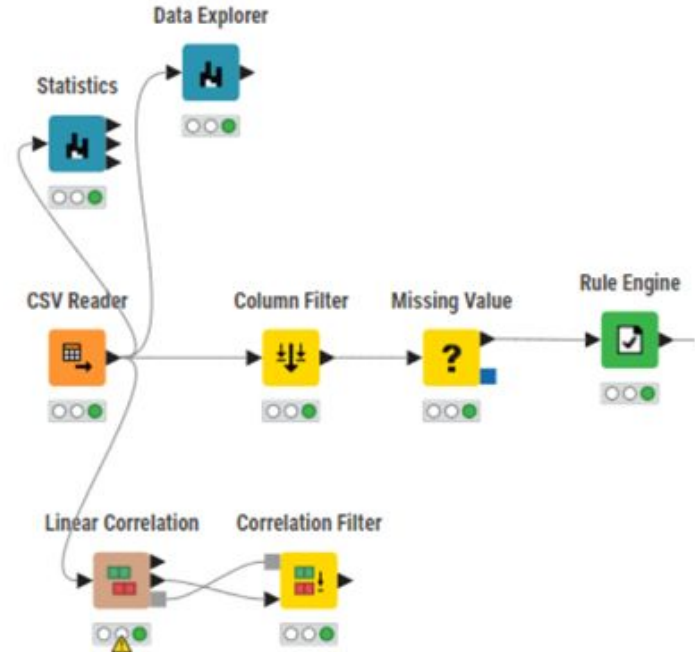
Data Preparation – Missing Values

- Open the missing value's configuration panel.
- For both numeric attributes, such as double and integer type, choose median.

Number (integer)	Median
String	Do nothing
Number (double)	Median

Data Preparation – Label Encoding

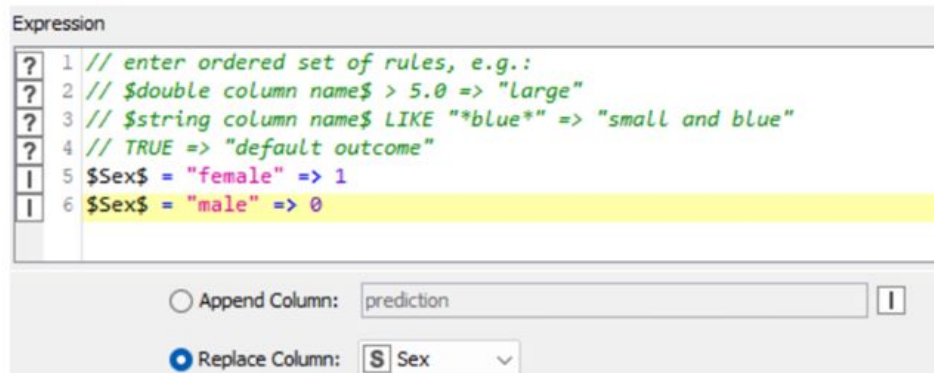
- Label encoding is a technique used in Machine Learning to convert categorical data into numerical form.
- In the Node Repository, search for the Rule Engine node. Drag and drop it into the Workflow Editor. Connect the output port of the Column Filter node to the input port of the Rule Engine node.



Data Preparation – Label Encoding

- Open the Rule Engine node's configuration panel, and, under Expression, write the rules that will map the categorical values to numerical values:

```
$Sex$ = "female" => 1  
$Sex$ = "male" => 0
```



Data Preparation – Label Encoding

- Select the option Replace Column and choose the column Sex.
- Click OK and execute the node.
- Inspect the output in the node monitor. You should see that the values of the attribute have changed.

☐ Append Column: prediction

☒ Replace Column:

Sex
Number (integer)
0
1
1
1
0

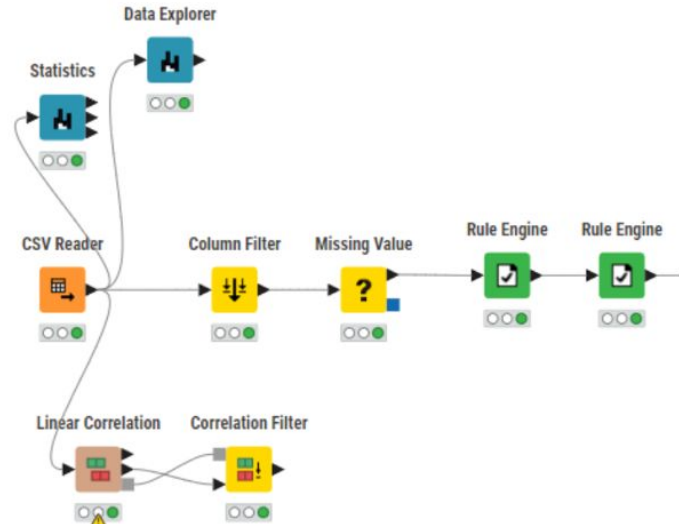
Data Preparation – One Hot Encoding

- One-hot encoding allows categorical features to be represented numerically, but without implying any ordinal relationship between categories.
- It creates separate columns for each category without imposing any order or magnitude on them.

Data Preparation – One Hot Encoding

Let's apply it to the the Pclass attribute:

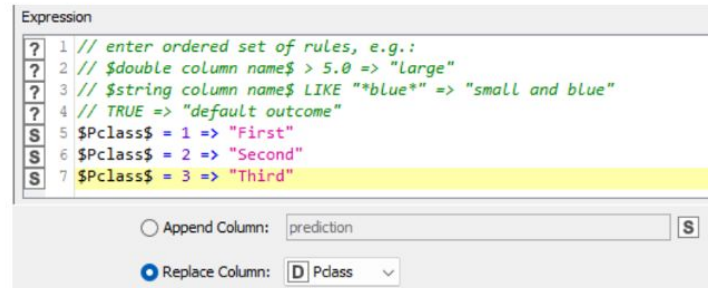
- First, rename the values for better readability and interpretation. Instead of keeping the numeric values (1, 2, 3), convert them to more meaningful labels: 1 to "First", 2 to "Second" and 3 to "Third".
- In the Node Repository, search for the Rule Engine node. Drag and drop it into the Workflow Editor. Connect the output port of the last Rule Engine node to the input port of the new Rule Engine node.



Data Preparation – One Hot Encoding

- Open the Rule Engine node's configuration panel, and, under Expression, write the following rules:

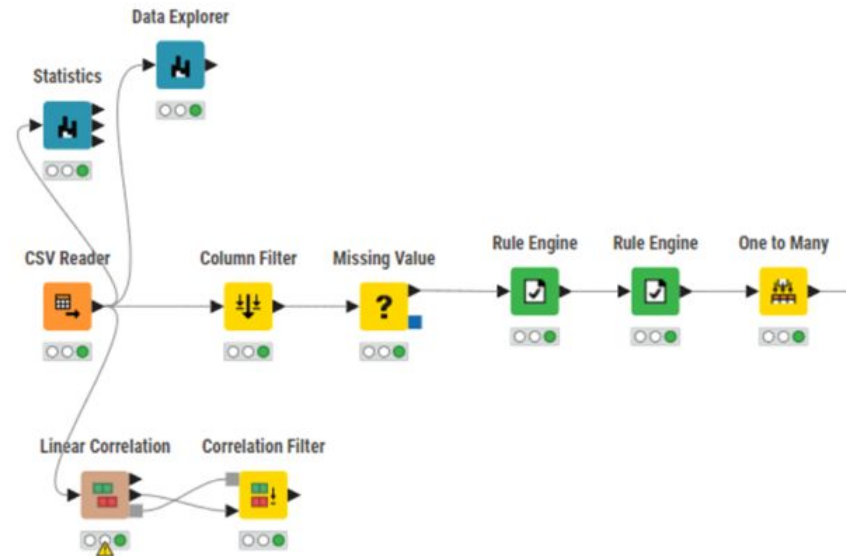
```
$Pclass$ = 1 => "First"  
$Pclass$ = 2 => "Second"  
$Pclass$ = 3 => "Third"
```



- Make sure to select Replace Column. Click Ok and run the node.

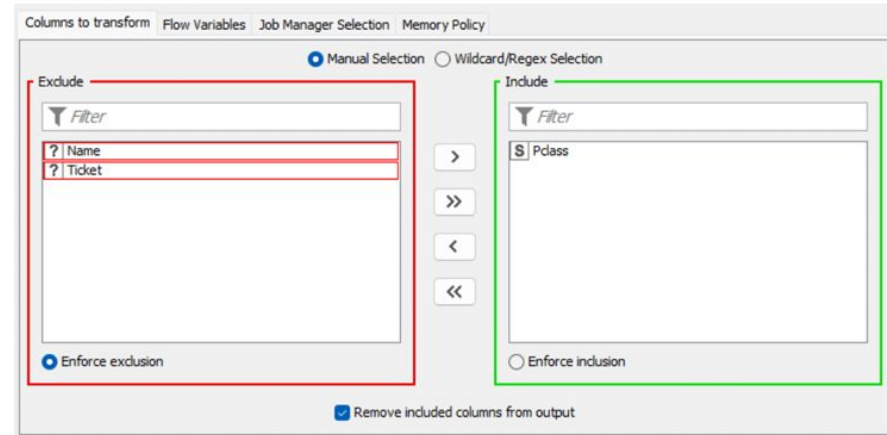
Data Preparation – One To Many

- In the Node Repository, search for the One to Many node. Drag and drop it into the Workflow Editor. Connect the output port of the last Rule Engine node to the input port of the One to Many node.
- The One to Many node in KNIME is used to apply one-hot encoding to categorical variables, transforming them into binary (0/1) columns.



Data Preparation – One To Many

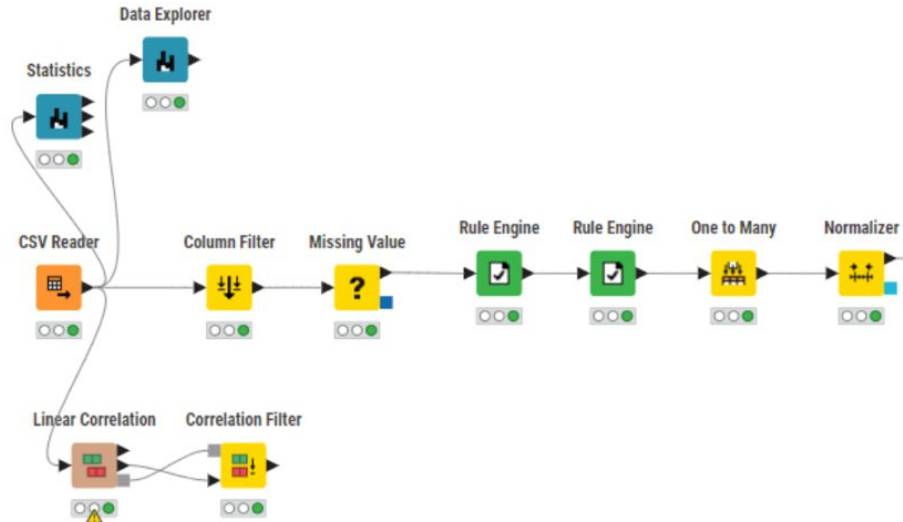
- Open the node's configuration panel. Make sure to only include the **Pclass** feature. Also, select the option remove included columns from output so that the old **Pclass** column is excluded.
- Click Ok and run the node. In the output monitor, you can see the addition of three new variables, one for each class.



Third Number (integer) ▾	First Number (integer) ▾	Second Number (integer) ▾
1	0	0
0	1	0
1	0	0
0	1	0
1	0	0
1	0	0
0	1	0
1	0	0

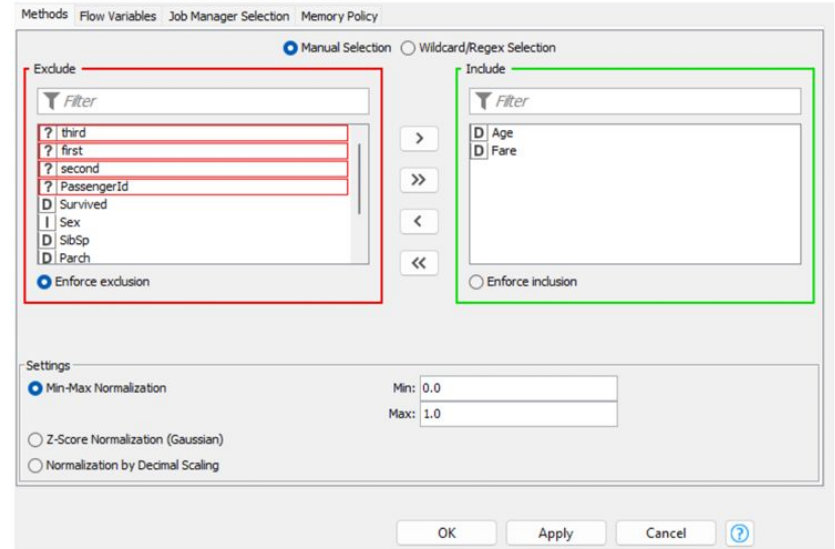
Data Preparation – Feature Engineering

- In the Node Repository, search for the Normalizer node. Drag and drop it. Connect the output port of the One to Many node to the input port of the Normalizer node.



Data Preparation – Feature Engineering

- Open the node's configuration panel. Make sure to only include the Age and Fare attribute. Additionally, choose Min-Max as normalization method. This method rescales the values so they fall between a specified minimum and maximum, between 0 and 1.



Methods | Flow Variables | Job Manager Selection | Memory Policy

☒ Manual Selection ☐ Wildcard/Regex Selection

Exclude

Filter

- ? third
- ? first
- ? second
- ? PassengerId
- D Survived
- I Sex
- D SibSp
- D Parch

☒ Enforce exclusion

Include

Filter

- D Age
- D Fare

☐ Enforce inclusion

Settings

☒ Min-Max Normalization Min: 0.0
Max: 1.0

☐ Z-Score Normalization (Gaussian)

☐ Normalization by Decimal Scaling

OK Apply Cancel ?

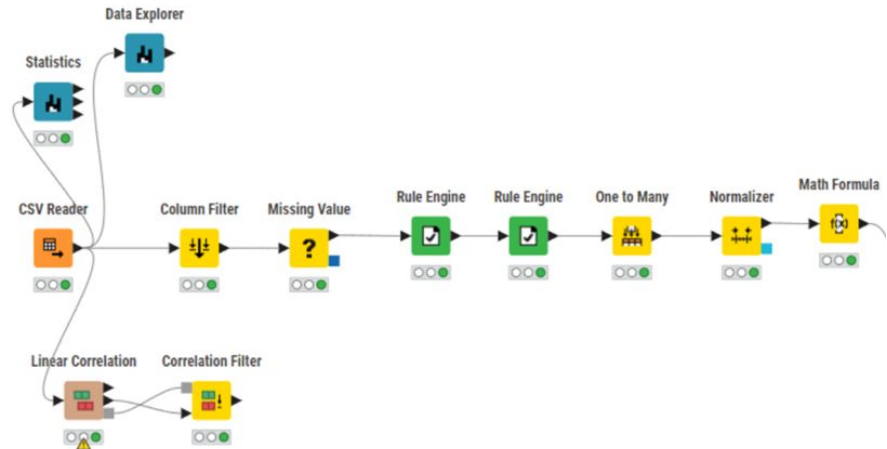
Data Preparation – Feature Engineering

- Click Ok and run the node to see the modified values. You can see that the values are between 0 and 1, e.g., normalized.

Age Number (double)	SibSp Number (double)	Parch Number (double)	Fare Number (double)
0.271	1	0	0.014
0.472	1	0	0.139
0.321	0	0	0.015
0.435	1	0	0.104
0.435	0	0	0.016
0.347	0	0	0.017
0.673	0	0	0.101

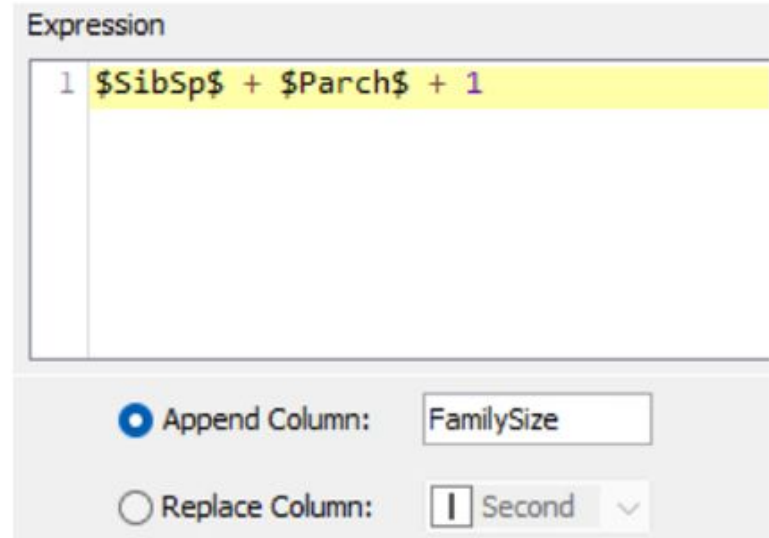
Data Preparation – Adding new variables

- Adding new variables in a clustering process can offer several benefits that improve the quality of the analysis and the effectiveness of the model.
- In the Node Repository, search for the Math Formula node. Drag and drop it. Connect the output port of the previous Math Formula node to the input port of the new Math Formula node.



Data Preparation – Adding new variables

- On the node's configuration panel, write the following expression: `$SibSp$ + $Parch$ + 1`. Choose the Append Column option and write "FamilySize".
- We are creating a new variable called FamilySize by summing the number of siblings/spouses (SibSp) and the number of parents/children (Parch) aboard. Additionally, we add 1 to account for the passenger themselves.



The screenshot shows the configuration panel for a KNIME node. At the top, the title "Expression" is displayed. Below it, a text area contains the expression `1 $SibSp$ + $Parch$ + 1`, which is highlighted in yellow. At the bottom, there are two radio button options: "Append Column:" (which is selected) and "Replace Column:". The "Append Column:" option has a text box next to it containing the name "FamilySize". The "Replace Column:" option has a dropdown menu next to it showing "Second" with a downward arrow.

Data Preparation – Adding new variables

- Run the node so you can see the new variable showing up.
- The data is now prepared and ready for training a K-Means model.

Second <i>Number (integers)</i>	FamilySize <i>Number (double)</i>
0	2
0	2
0	1
0	2
0	1

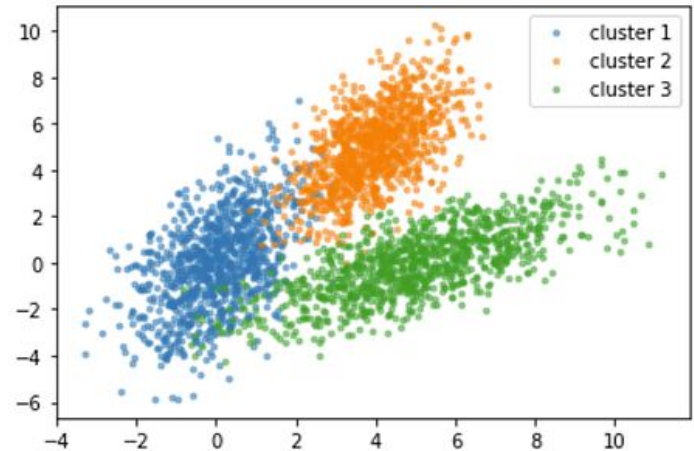
Clustering

Clustering is a Machine Learning technique that **groups objects, or data points**, based on **similarities**. These groups are represented on scatter plots for data visualization, prototyping, sampling and segmentation.

Clusters

A **clustering** algorithm takes a **large amount of unlabeled input data** to identify any possible groupings within the data. We refer to those collections as **clusters**.

A cluster is a **collection of data points** that share similarities with one another as a result of how they relate to other data points.



A scatter plot with different clusters denoted by different colours

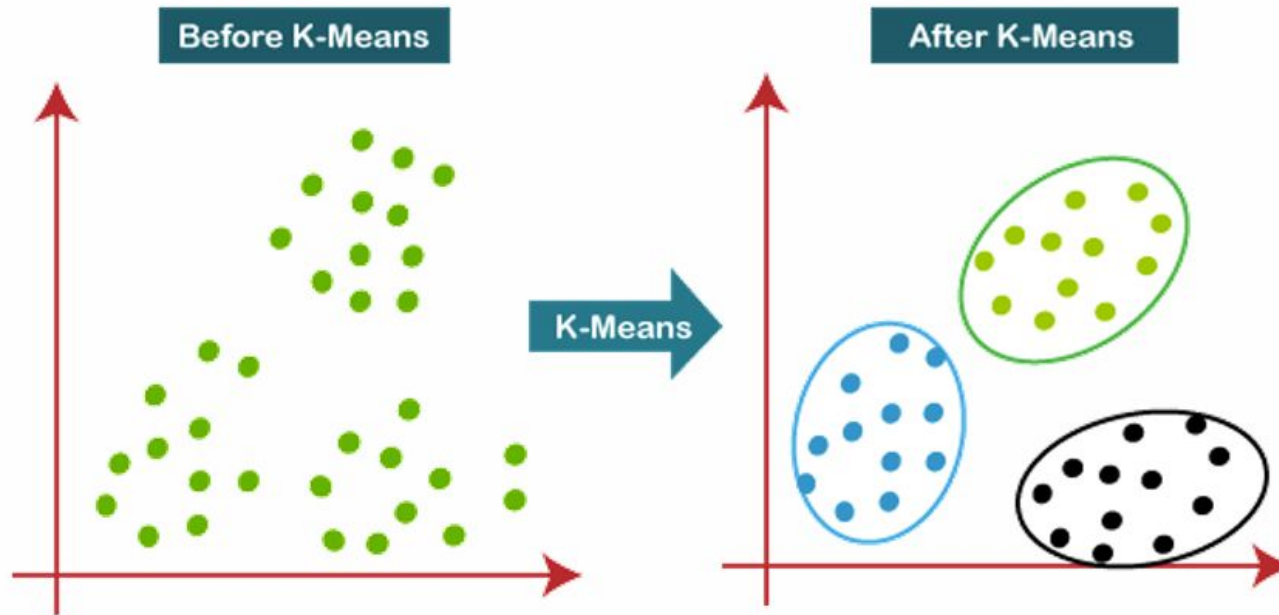
K-Means

- Most frequently utilized clustering algorithm;
- **Unsupervised learning algorithm** that relies on **centroids**;
- Centroids represent the **"center"** or average position of all data points in a cluster;
- Aims to **reduce the variability of data points** inside a cluster;
- Most effective when applied to small datasets.

K-Means

- Applies a **partitioning technique** that divides data space into distinct clusters;
- Randomly chooses **cluster centers** and assigns all data points to the closest cluster centers;
- The cluster centers are **recalculated** to be the centroids of the clusters that have been newly created.
- The datapoint sare moved to the **closest cluster** centers that were recently recalculated.
- The process is **repeated** until the cluster centers each a stationary position.

K-Means

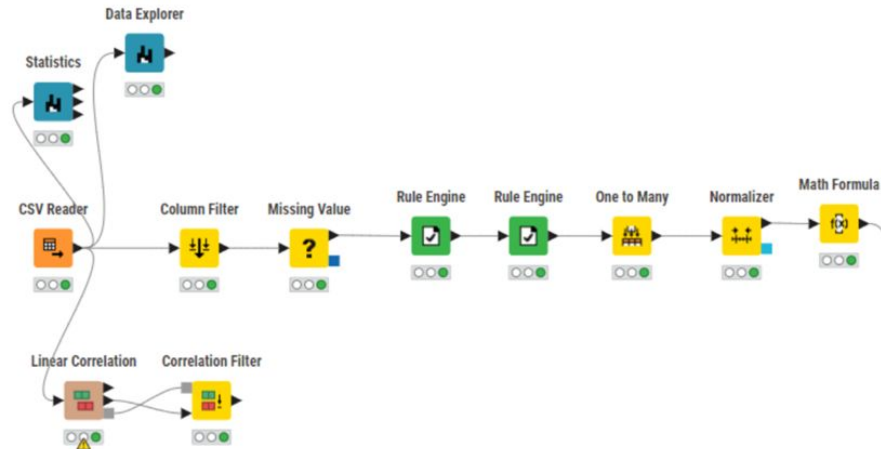


Exercise

The objective is to **group passengers into clusters** based on **similarities** in their features, **without using the survival information as a label**. Instead of predicting survival, the goal is to identify natural groupings of passengers that share similar characteristics.

Exercise

Pre-processed workflow



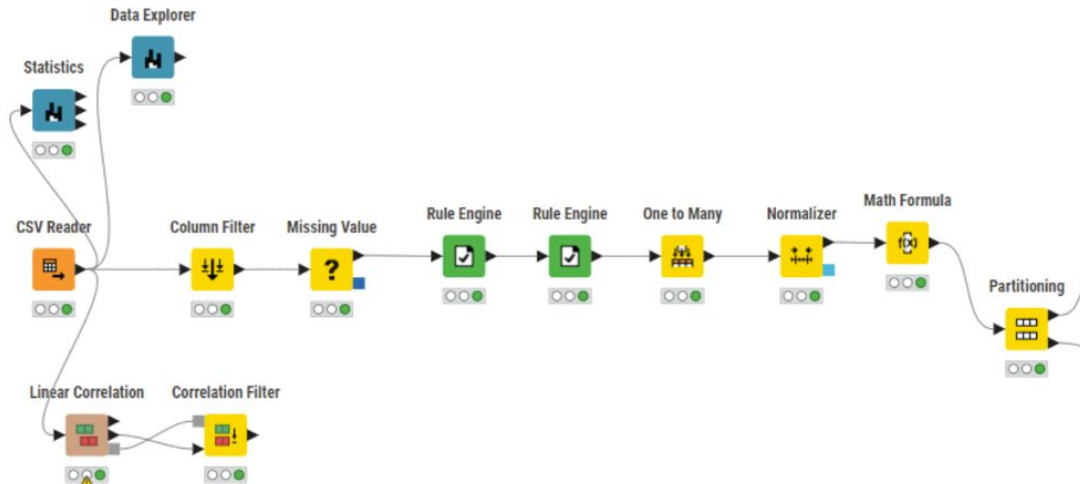
Partitioning

Partitioning the data into **training and testing sets** is a fundamental step in model development;

- **Train a model on one subset** of the data (training set);
- **Evaluate** how well the trained model **generalizes to unseen data** (test set).

Partitioning

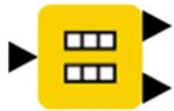
- In the Node Repository, search for the **Partitioning** node. Drag and drop it. Connect the **output port** of the **Math Formula** node to the **input port** of the **Partitioning** node.



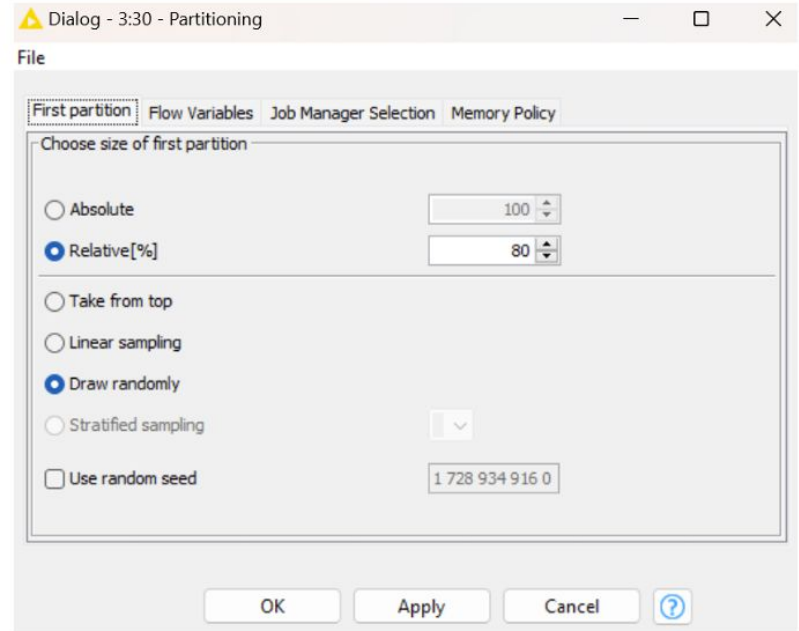
Partitioning

- Open the node's configuration panel.
Write 80 on the Relative(%) option.
Apply the changes and run the node.

Node: Partitioning



*Splits a dataset into
training and testing subsets*



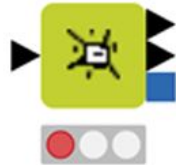
K-means Node

- Apply the K-Means node to train the unsupervised model.
- In the Node Repository, search for the k-Means node. Drag and drop it. Connect one of the output ports of the Partitioning node to the input port of the k-Means node. Run the node.



Train the Model

- Open the node's configuration panel. Start with a number of clusters equal to 2 and make sure to exclude the target variable, e.g., Survived.



*Trains a K-Means
model*

Clusters

Number of clusters: 

Centroid initialization:

☐ First k rows

☒ Random initialization ☒ Use static random seed

Number of Iterations

Max. number of iterations: 

Column Selection

Exclude

 Filter

☒ Survived

>

>>

<

<<

Include

 Filter

☐ Sex

☒ Age

☒ SibSp

☒ Parch

☒ Fare

☐ Third

☐ First

☐ Second

☒ FamilySize

☐ Always include all columns

Train the Model

- You can now see a new variable, called Cluster, that contains all the generated clusters. In this case, 3 clusters were created.

Second Number (inte... 	FamilySize Number (dou... 	Cluster String
0	2	cluster_0
0	1	cluster_0
0	2	cluster_1
0	1	cluster_1
0	3	cluster_2

Cluster Assigner

- Apply the Cluster Assigner node to infer on the test data using the model trained on the k-Means node.
- In the Node Repository, search for the Cluster Assigner node. Drag and drop it. Connect the second output port of the Partitioning node to one of the input ports of the Cluster Assigner node. Connect the other input port to the K-Means node. Run the node



- Assigns new data points to the nearest cluster based on an existing clustering model;
- Allows users to classify data according to predefined clusters

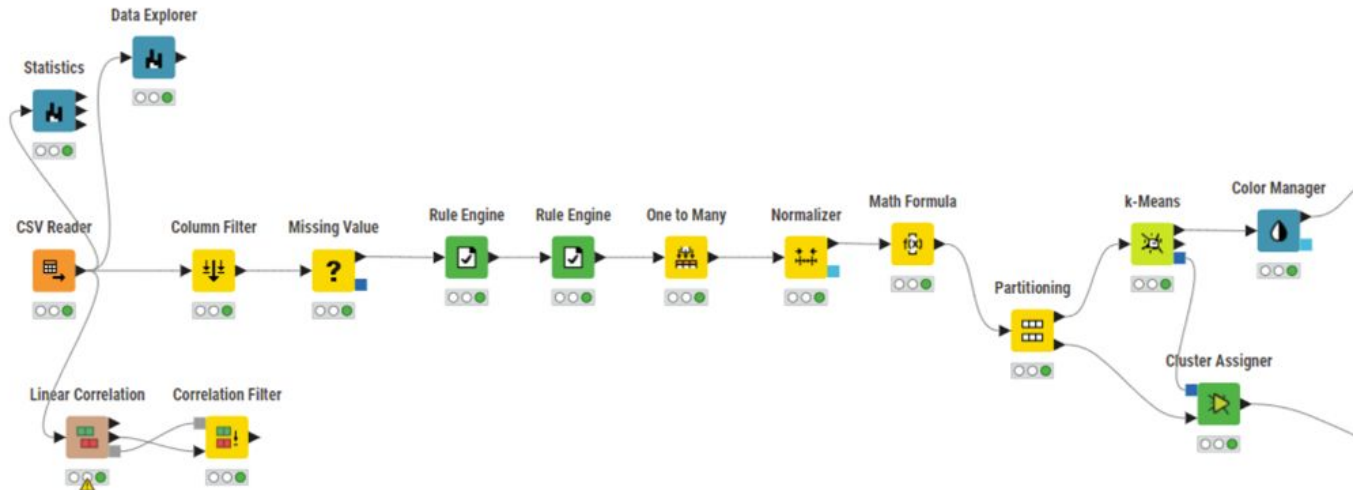
Cluster Assigner

- The Cluster Assigner node in KNIME is used to assign new data points to the nearest cluster based on an existing clustering model, allowing users to classify data according to predefined clusters.



Color Manager

- In the Node Repository, search for the Color Manager node. Drag and drop it. Connect the output port of the K-Means node to the input port of the Color Manager node.

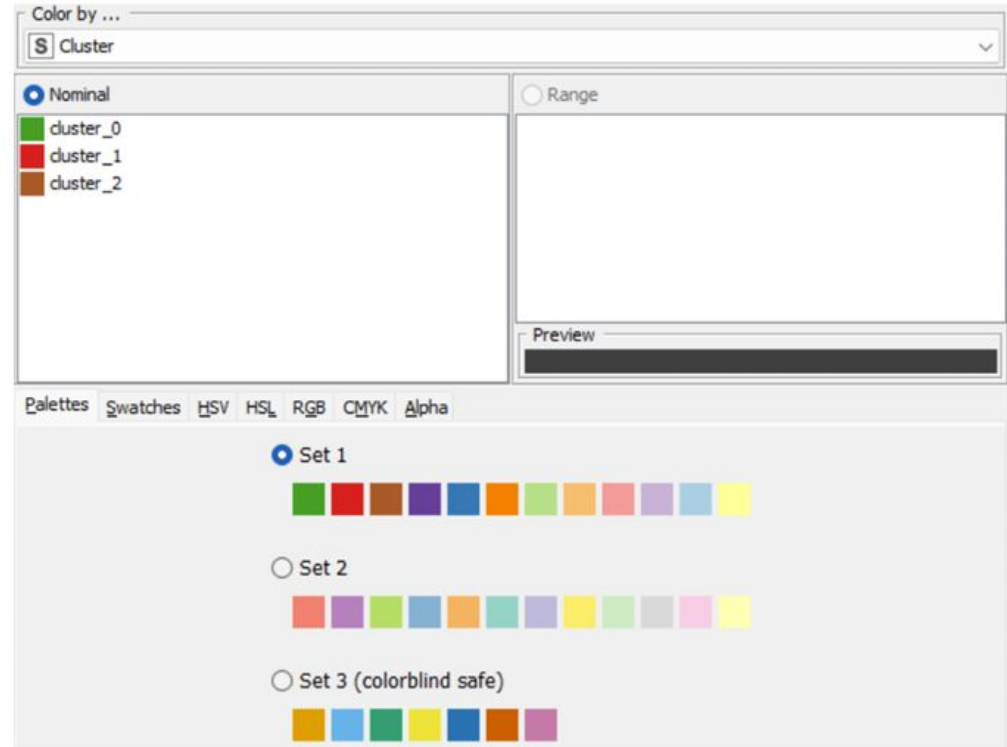


Color Manager

Node: Color Manager

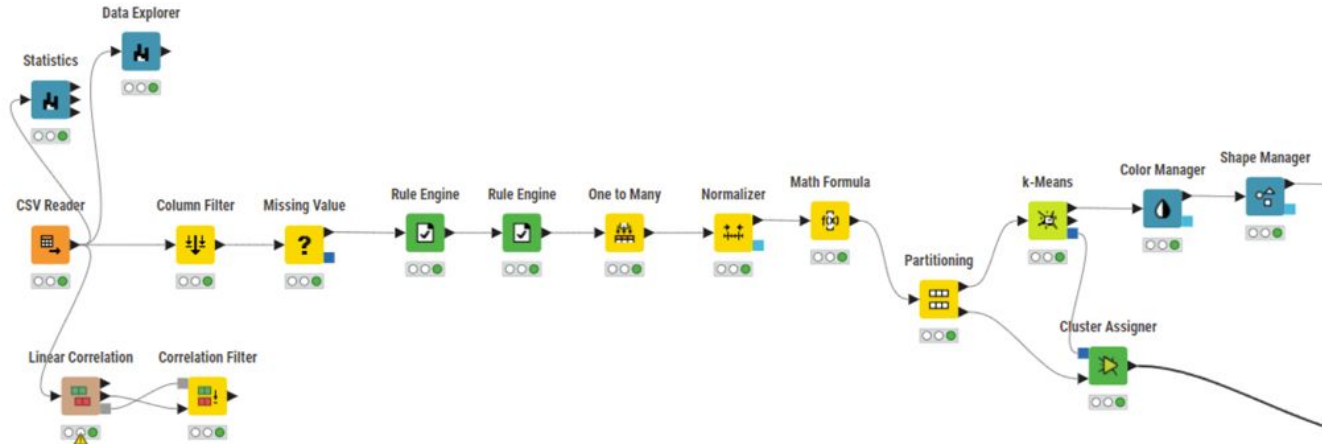


*Allows users to assign
specific colors to data
values.*



Shape Manager

- In the Node Repository, search for the Shape Manager node. Drag and drop it. Connect the output port of the Color Manager node to the input port of the Shape Manager node.



Shape Manager

Node: Shape Manager



Allows users to assign specific shapes to data values.

Shape Settings
 Flow Variables
 Job Manager Selection
 Memory Policy

Select nominal column:

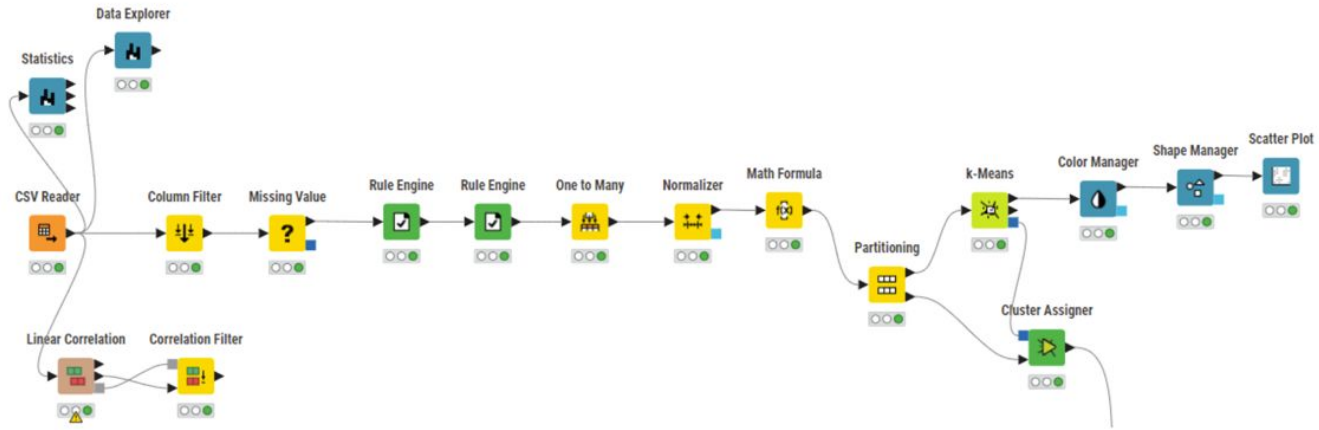
S Cluster
 ▼

Shape Mapping

Values of Cluster	Shapes
cluster_0	■ Rectangle
cluster_1	● Circle
cluster_2	▲ Triangle

Scatter Plot

- In the Node Repository, search for the Scatter Plot node. Drag and drop it. Connect the output port of the Shape Manager node to the input port of the Scatter Plot node.



Scatter Plot

- Analyze the different features within the Scatter Plot node's configuration panel to gain insights.
- Open the Scatter Plot node's configuration panel. For the horizontal dimension choose Age and for the vertical dimension choose Fare. For the Color Dimension choose Cluster.



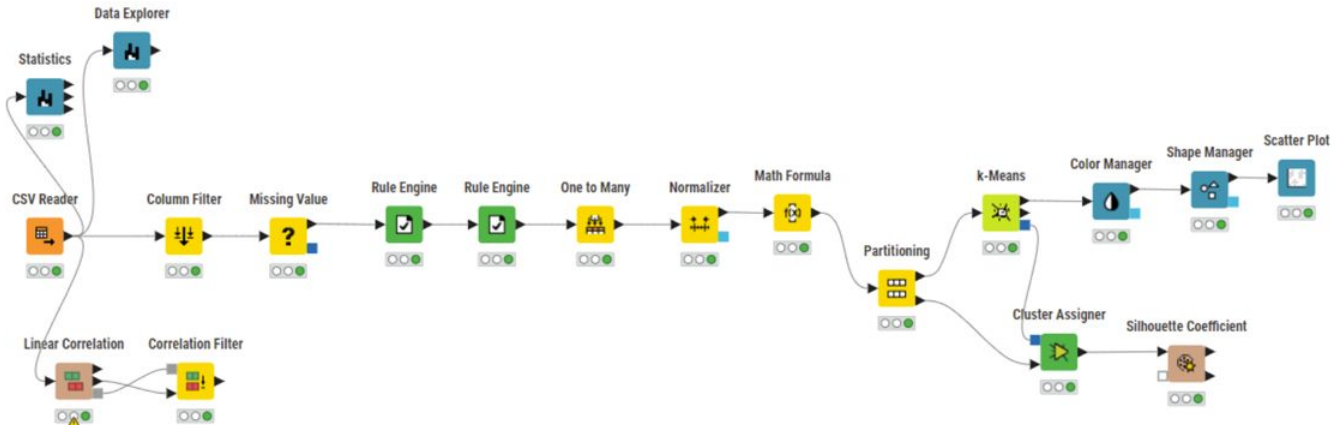
Scatter Plot

- Keep the horizontal dimension as Age and choose Family Size as the Vertical Dimension.



Evaluate the model

- In the Node Repository, search for the Silhouette Coefficient node. Drag and drop it. Connect the output port of the Cluster Assigner node to the input port of the Silhouette Coefficient node.



Evaluate the Model

Measure:

Silhouette Score



- Assesses a dataset's well-defined clusters. The cohesiveness and separation between clusters are quantified.
- Better-defined clusters are indicated by higher scores, which range from -1 to 1.
- An object is well-matched to its own cluster if its score is close to 1.
- A score of about -1 suggests that the object might be in the incorrect cluster.

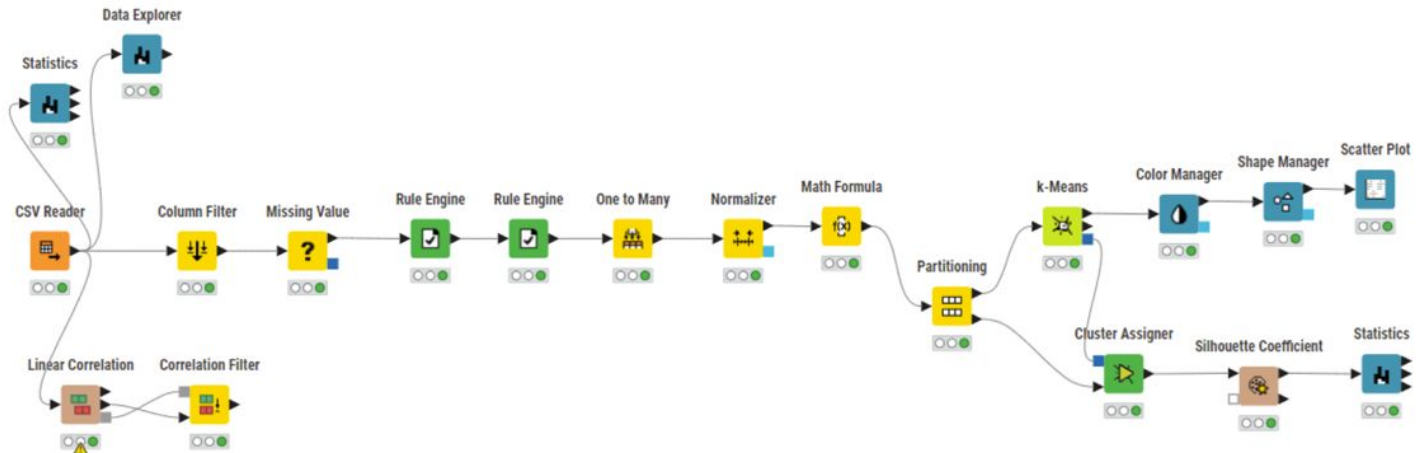
Evaluate the Model

- Run the node and verify the results.

FamilySize <i>Number (dou...</i>	Cluster <i>String</i>	Silhouett... <i>Number (dou...</i>
2	cluster_1	0.528
1	cluster_1	0.77
1	cluster_1	0.773
5	cluster_0	0.302
1	cluster_1	0.769
1	cluster_1	0.639
1	cluster_1	0.689

Evaluate the Model

- In order to effectively analyze the node's results, in the Node Repository, search for the Statistics node. Connect the output port of the Silhouette Score node to the input port of the Statistics node.



Evaluate the Model

- Run the node and click on its magnifying glass icon.

FamilySize	1	2,0391	?	11
Silhouette Coefficient	-0,2626	0,5417	?	0,7729

Evaluate the Model

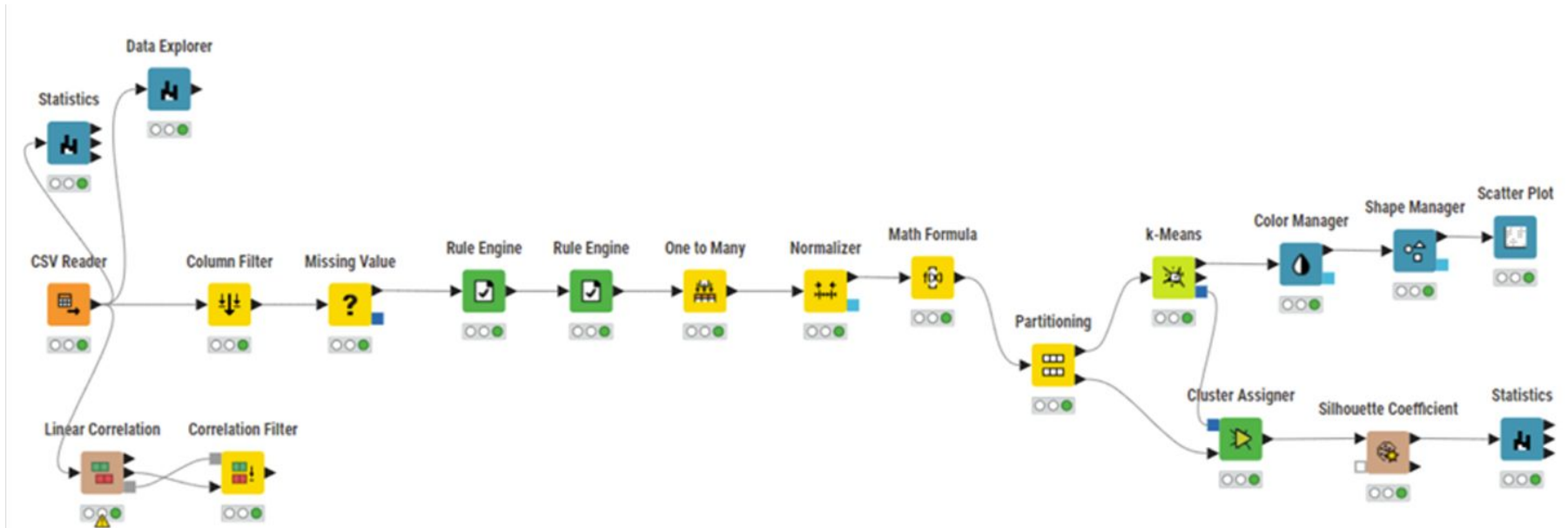
- **Maximum score = 0.7729.** Highly positive value, the corresponding point is well clustered.
- **Minimum score = -0.2626.** The corresponding point is likely positioned near the boundary between two clusters and may be ambiguous or poorly clustered.

Silhouette Coefficient	-0,2626	0,5417	?	0,7729
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Evaluate the Model

- The presence of both high and slightly negative scores suggests that, while the clustering model generally performs well for many points, there are some points for which the model **struggles to clearly distinguish**.
- Could be related to **overlapping cluster boundaries** or **complex patterns** in the data.
- Further tuning might be necessary for improved results.

Final Workflow





Universidade do Minho
Escola de Engenharia
Departamento de Informática

KNIME

Titanic Dataset

MESTRADO EM ENGENHARIA DE TELECOMUNICAÇÕES E
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